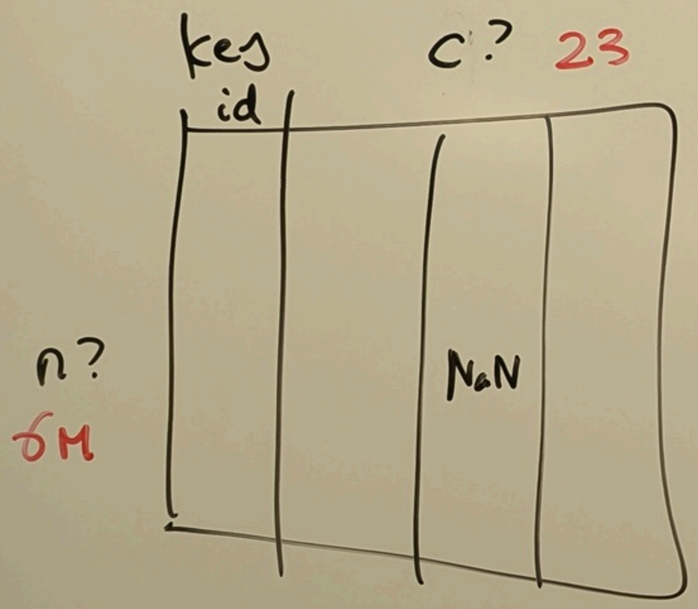
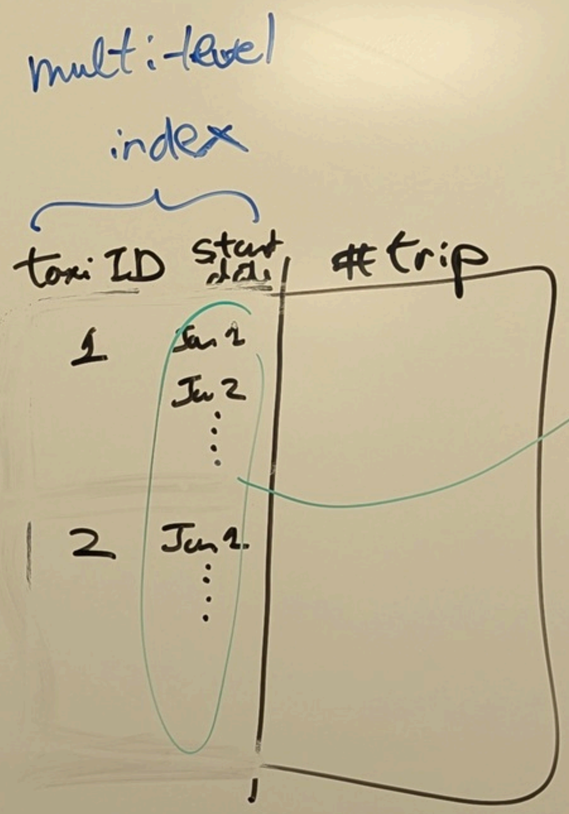
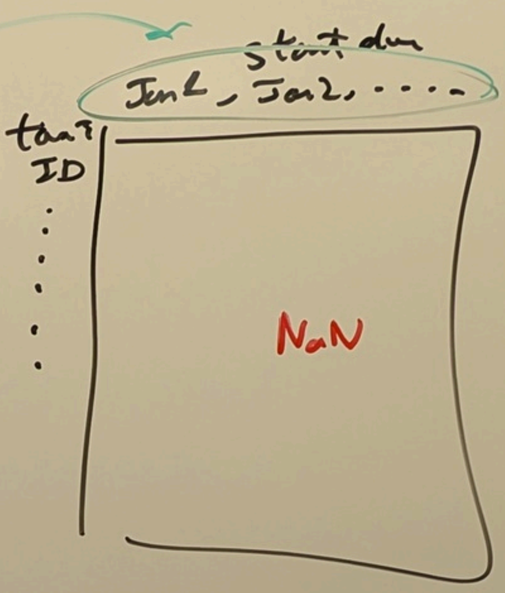
Back at 7:50pm



n?
δM



unstack



measure linear relationship

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{SD(X)\cdot SD(Y)}$$

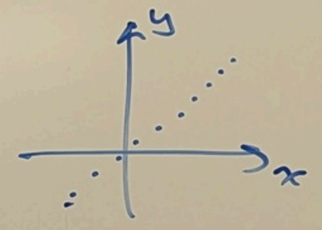
+? -?

$-1 < \ldots < 1$

"linear"

$y = f(x)$





$f(ax) = af(x)$

$f(x+b) = f(x) + b$

$\Rightarrow f(ax+b) = af(x) + b$

linear / affine transformation

$\text{Corr}(aX+b, cY+d)$

$=$

$\text{Corr}(X,Y)$

$$\text{Var}(X) = E\left(\left(X - E(X)\right)^2\right)$$

avg (distance)^2

$-\infty < \text{Cov}(X,Y) < \infty$

$$\text{Cov}(X,Y) = E\left(X - E(X)\right)\left(Y - E(Y)\right)$$

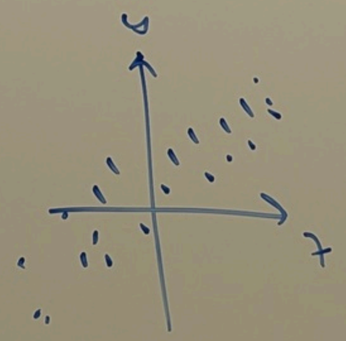
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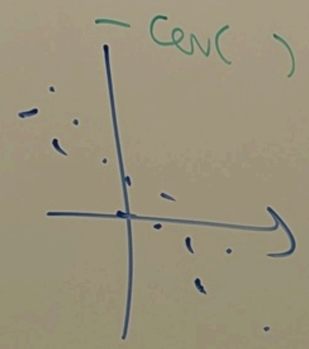
<0

>0 >0

<0 >0

>0 <0





n data points

$(\vec{x}_1, y_1), \ldots, (x_n, y_n)$

$(x_{11}, \ldots, x_{1d}) \in \mathbb{R}^d$

"belongs to"

$\in \mathbb{R}$ 0.6

$\in \{c_1, \ldots, c_k\}$

coin tosses

e.g. $x_1, \ldots x_n \sim \text{Bernoulli}(p)$ (unknown)

$\cap$

$\{H, T\}$

$\| \quad \|$

$1 \quad 0$

parameter

$P(\text{head}) = p$

$P(\text{tail}) = 1-p$

$p_{true} = 0.49$

$y = f(x)$

relationship

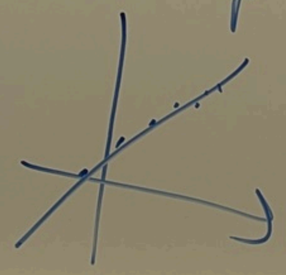
f = linear regression

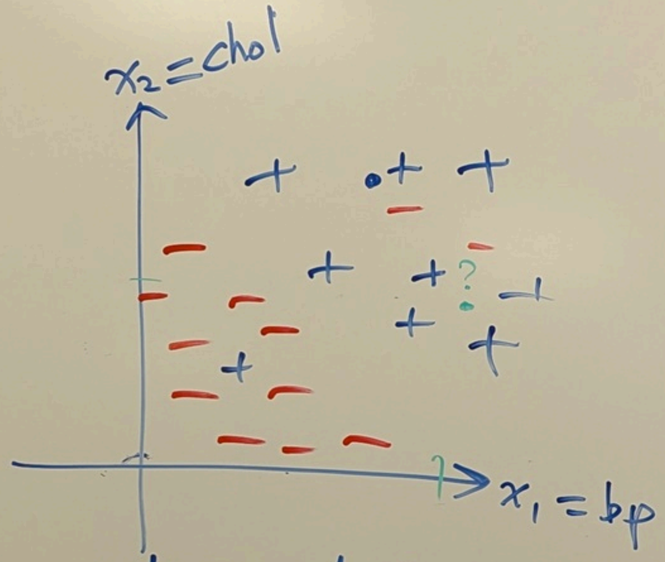
= neural net

e.g. $y = ax+b$

$y = \beta_0 + \beta_1 x + \varepsilon$

intercept slope





non-parametric model:

complexity of model ↑ as n ↑

parametric model:

params = fixed as n ↑

e.g. $\vec{x} = (bp, BMI)$

x_1 x_2

y = chol level e.g. 5.2

$(chol) = \overset{?}{f}(bp, bmI)$

neural net — linear regression

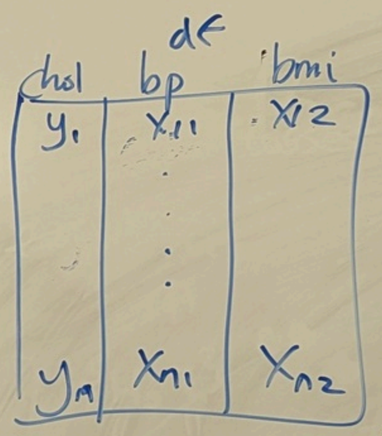
5.5 $= \beta_0 + \beta_1(\overset{140}{bp}) + \beta_2(\overset{30}{bmI}) + \varepsilon$

linear relationship

avg — individual error

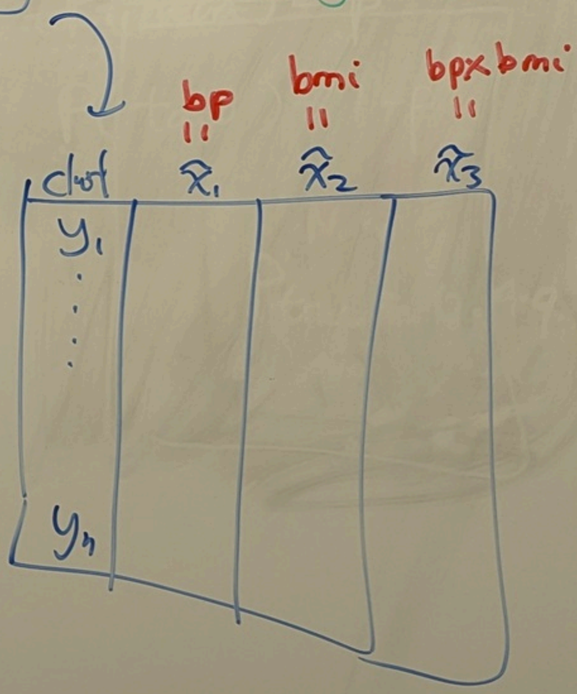
$$chol = 2.5 + 5(bp) + 20(bmi)$$

+20 +0 +1

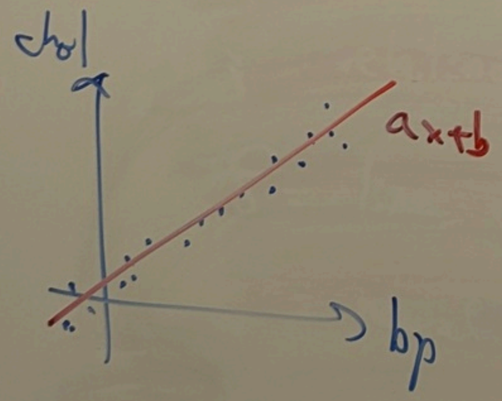


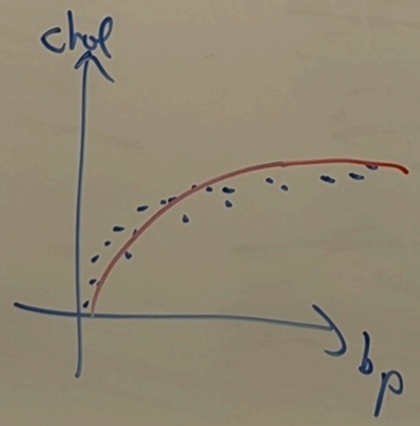
feature engineering

$$chol = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}}(bp) + \beta_2(bmi) + \beta_3(bp \times bmi)$$



$$chol = \beta_0 + \beta_1(\hat{x}_1) + \beta_2(\hat{x}_2) + \beta_3(\hat{x}_3)$$





$chol \approx \beta_0 + \beta_1 \log(bp)$

$chol \approx \beta_0 + \beta_1 \sqrt{bp}$

